

Shirui (Ray) WANG

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An enthusiastic and responsible educator with three years of undergraduate-level math and statistics teaching experience, dedicated to fostering effective learning environments and supporting student success in the age of AI.

Education

Ph.D. in Biostatistics	Jan 2020 – May 2026 (expected)
Georgia State University	Atlanta, GA, US
M.S. in Mathematics	Aug 2016 – June 2018
University of California, Irvine	Irvine, CA, US
B.S. in Applied Mathematics	Aug 2012 – June 2016
China Agricultural University	Beijing, China

Work Experience

Georgia State University	Atlanta, GA
Graduate Teaching Assistant (Instructor)	Aug 2022 – May 2025
- Taught elementary statistics as primary instructor to non-statistics major undergraduates, effectively conveying statistical concepts and approaches to students from diverse backgrounds - Earned an average of 4.5/5 on the student evaluation and 4.4/5 on RateMyProfessors.com - Won the Best Graduate Teaching Assistant Award (2025) for outstanding teaching performance	
Graduate Lab Assistant	Jan 2020 – Jul 2022
Assisted instructors with grading and tutoring in Probability, College Algebra and Precalculus class	
American Cancer Society	Atlanta, GA
Biostatistics Research Intern, Cancer Risk Prediction and Health Disparities	May 2025 – Nov 2025
- Developed imputation frameworks based on EM algorithm and sequential quadratic programming (SQP) algorithm, integrated with flexible parametric survival models, to address missing stage-at-diagnosis data in SEER datasets - Performed a multinomial logistic regression analysis to study the association between Medicaid coverage continuity and cancer stage at diagnosis, finding that continuous coverage significantly reduced the likelihood of distant cancer	
SUNY Upstate University Hospital	Remote
Biostatistician Intern	Dec 2023 – Jan 2024
- Conducted a retrospective analysis of sepsis outcomes from a 2^2 factorial observational study to evaluate the impact of orderset and note utilization on key healthcare outcomes (mortality rate, length of stay, ICU utilization, etc.) - Performed Chi-square tests, Friedman tests, Mixed Effects Models, and Poisson Regression with risk-adjustment offsets, providing robust evidence that orderset and note utilization significantly reduced mortality and ICU utilization.	
Georgia State University	Atlanta, GA
Doctoral Researcher	Aug 2021 – Dec 2025
- Developed novel statistical approaches for evaluating continuous diagnostic tests (ROC, AUC, Youden index) with partially missing gold standard verification using Empirical Likelihood and bootstrap methods - Conducted extensive Monte Carlo simulation studies and performed biomarker validation analyses for real-world cancer and AD datasets in R, achieving accurate and robust inferences when only 36% of the subjects were verified Developed an R package <i>rocub</i>, integrating newly proposed statistical methods into an accessible toolkit used by practitioners and non-technical users	
China Construction Bank	Shenyang, China
Data Mining Intern	July 2019 – Aug 2019
- Developed predictive models for retail banking operations, training a Random Forest Classifier in Python to assess customer responsiveness to telemarketing, achieving 93% accuracy on the testing set - Conducted risk analysis on transaction data to detect potential financial crimes, identifying 500+ gambling accounts based on transaction patterns	

Technical Skills

- **Programming and Computing:** **R** (Package, ggplot2, Shiny, Markdown, Parallel Computing), **SAS**, **Python** (Numpy, Pandas), MATLAB, SQL, C, High-performance computing clusters, JMP
- **Statistics: Regression Analysis** (GLM, Lasso/Ridge regression), **Longitudinal Data** (Mixed effects model, Repeated Measures), **Survival Analysis** (Cox model, censored data), **Missing Data** (Multiple Imputation, Propensity scores, EM, Inverse Probability Weighting), **Real-World Data** (SEER, Medicaid), **Computational Methods** (MCMC, Bootstrap), **Clinical Trials** (Sample size calculation, Power analysis, Multiplicity adjustments, Experimental Design), **Diagnostic and Biomarker studies** (ROC, AUC), Bayesian Methods, Multivariate Statistics, Meta analysis, Causal Inference
- **Machine Learning & Deep Learning:** Random Forest, XGBoost, LightGBM, SVM, K-Means, PCA, Deep Neural Networks, RNNs, GANs, LSTM, Diffusion Models, Large Language Models, Scikit-Learn, PyTorch

Publications & Working Papers

- **Wang S**, Shi S, Qin G. Interval estimation for the Youden index of a continuous diagnostic test with verification biased data. *Statistical Methods in Medical Research* 2025; 34: 796-811. doi:10.1177/09622802251322989
- Shi S, **Wang S**, Qin G. Interval estimation for three-class Youden index with verification bias. *Journal of Biopharmaceutical Statistics* 2025; 1: 1-22. doi:10.1080/10543406.2025.2549361
- **Wang S**, Shi S, Qin G. Empirical likelihood inference for the area under the ROC curve with verification biased data. *Revision at Statistical Methods in Medical Research*
- Shi S, **Wang S**, Qin G. Interval Estimation for the Sensitivity of a Test to the Early Diseased Stage with Verification Bias. *Revision at Pharmaceutical Statistics*
- Jia S, **Wang S**, Qin G. Methodological Approaches for the Estimation of Confidence Intervals on Partial Youden Index under Verification Bias. *Revision at Pharmaceutical Statistics*
- **Wang S**, Qin G. Empirical likelihood-based confidence intervals for sensitivity of a continuous test at a fixed level of specificity with verification bias. *Submitted*
- Jia S, **Wang S**, Qin G. Empirical Likelihood-Based Confidence Intervals for the Partial AUC with Verification Bias. *Submitted*
- **Wang S**, Islami F, Siegel RL, Jemal A, Choudhury PP. Imputation of missing cancer stage at diagnosis accounting for stage-specific survival. *Manuscript in preparation*
- Choudhury PP, Zhao J, **Wang S**, Jemal A, Yabroff R, Islami F. Are disruptions in Medicaid coverage linked to advanced stage diagnosis of screen-detectable cancers? *Manuscript in preparation*

Conferences & Presentations

- 2025 Joint Statistical Meeting (JSM), Contributed Paper Presentation, Nashville, TN, Aug 2025
- 2025 Georgia Statistics Day, Student Poster, University of Georgia, Athens, GA, Oct 2025
- The 10th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, GA, May 2025
- The Duke Industry Statistics Symposium (DISS), Duke University, Durham, NC, April 2025
- 3rd Annual Graduate Conference on Research, Scholarship & Creative Activity, Student Poster, Georgia State University, Atlanta, GA, Feb 2025
- The 9th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, GA, May 2024
- Scientific Computing Day, Student Poster & Lightning Talk Session, Georgia State University, Atlanta, GA, Nov 2023

Awards

- Wiley M. Suttles Math Award for outstanding math graduate student, Georgia State University, 2025
- Graduate Scholar Travel Award, Georgia State University, 2025
- Best Graduate Teaching Assistant Award for outstanding teaching performance, Georgia State University, 2025
- Mathematics Achievement Award, Georgia State University, 2022
- Best undergraduate theses (4/60), Department of Mathematics, China Agricultural University, 2016
- First Prize (Beijing) in China Undergraduate Mathematical Contest in Modeling , China Agricultural University, 2014
- Second Prize (Beijing) in China Undergraduate Mathematical Contest in Modeling , China Agricultural University, 2013
- Third-class Scholarship for Outstanding Students of College , China Agricultural University, 2012 & 2014
- Blood donation volunteer & China Marrow Donor Program volunteer, 2012